Incorporating DRAG into Space-Curve Quantum Control and Leakage-Aware BARQ

Technical design note for the `qurveros` package

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Abstract

This note derives how Derivative Removal by Adiabatic Gate (DRAG) can be combined with space-curve quantum control (SCQC) and the Bézier ansatz for robust quantum control (BARQ), as implemented in `qurveros`. The central result is that a DRAG correction applied after a completed BARQ design is inexpensive, but it does not automatically preserve the exact geometric robustness conditions. It preserves the computational gate only perturbatively in $1/(\alpha T)$ and generally introduces mixed errors of order $(\delta_z T)^2/(\alpha T)^2$. A leakage-aware BARQ objective can instead impose four real first-order leakage constraints while recomputing dephasing and amplitude sensitivities under the corrected qutrit evolution. The note also records the associated boundary conditions, objective functions, spectral and smoothness extensions, the relation to a perturbative $\mathbb{C}^2$ leakage curve, and the scaling that can reasonably be expected before a numerical campaign.

Contents

1 Scope and conventions

SCQC and BARQ design a single-qubit gate by constructing a space curve whose curvature and torsion determine the drive. Those constructions are exact for a two-level Hamiltonian. A transmon, or any weakly anharmonic ladder, has at least one additional level. DRAG suppresses the resulting leakage by a near-identity frame transformation whose generator is proportional to the drive [**?**]. The two ideas therefore meet at the pulse, not inside the original three-dimensional Frenet frame.

Throughout, $P = |0\rangle\langle 0| + |1\rangle\langle 1|$ projects onto the computational subspace and $Q = I - P$. Time t runs from 0 to the gate duration T. The dimensionless time used by `qurveros` is

$$s = \frac{t}{T} \in [0, 1]. \tag{1}$$

A dot denotes $\mathrm{d}/\mathrm{d}t$ and a prime denotes $\mathrm{d}/\mathrm{d}s$.

The minimal physical model is a driven three-level ladder. In the frame rotating at the carrier frequency, and after the rotating-wave approximation,

$$\begin{aligned}\frac{H_3(t)}{\hbar} &= \delta(t)|1\rangle\langle 1| + (2\delta(t) + \alpha)|2\rangle\langle 2| \\ &\quad + \frac{1}{2}\Big\{\varepsilon(t)\big(|0\rangle\langle 1| + \lambda|1\rangle\langle 2|\big) + \text{h.c.}\Big\}.\end{aligned} \tag{2}$$

The complex envelope is $\varepsilon(t) = \Omega_x(t) + \mathrm{i}\Omega_y(t)$, λ is the ratio of the $1 \leftrightarrow 2$ and $0 \leftrightarrow 1$ couplings, and

$$\alpha = \omega_{12} - \omega_{01} \tag{3}$$

is the anharmonicity. For a transmon one typically has $\lambda = \sqrt{2}$ and $\alpha < 0$. Every quadrature sign below is tied to this convention. Taking the opposite complex-envelope convention, or replacing α by $|\alpha|$, flips signs but not the geometric conditions.

The `qurveros` control dictionary stores dimensionless fields. With $u(s)$ the complex envelope and $d(s)$ the detuning,

$$u(s) = T\varepsilon(Ts), \qquad d(s) = T\delta(Ts), \qquad \nu = \alpha T, \qquad \eta = \frac{1}{|\nu|}. \tag{4}$$

In the SCQC mapping, up to the package's sign conventions, $\omega(s) = T\kappa(t)$ is the normalized curvature, the phase or detuning is fixed by the torsion, and T is the length of the curve. DRAG is an asymptotic expansion in η. It is appropriate when the envelope and its phase vary slowly relative to $|\alpha|$, and it becomes progressively less reliable when $|\alpha|T$ is only a few radians.

Remark 1 (What is derived and what is a proxy)**.** Sections **??**, **??**, and **??** distinguish three objects: an exact spectral identity for a bare envelope, a first-order Dyson formula for gate leakage, and the original fixed-axis DRAG transformation. A complex-envelope formula for a rapidly varying SCQC phase is the natural extension of those results, but it remains an ansatz until it is checked by a qutrit Schrieffer–Wolff expansion or by direct simulation.

2 First-order leakage

Let $U_q(t)$ be the ideal evolution generated by the computational part of (**??**), embedded in the qutrit space, and let $\tilde{U}_q(s) = U_q(Ts)$. Treat the $1 \leftrightarrow 2$ coupling as a perturbation. For a computational input $|j\rangle$, the first-order amplitude to occupy $|2\rangle$ at time T is

$$A_{2j}^{(1)}(T) = -\mathrm{i}\frac{\lambda}{2}\int_0^T \mathrm{d}t\; \varepsilon^*(t)\, e^{\mathrm{i}\alpha t}\, \langle 1|U_q(t)j\rangle. \tag{5}$$

Changing to normalized variables cancels one factor of T:

$$A_{2j}^{(1)} = -\mathrm{i}\frac{\lambda}{2}\int_0^1 \mathrm{d}s\; u^*(s)\, e^{\mathrm{i}\nu s}\, \langle 1|\tilde{U}_q(s)j\rangle. \tag{6}$$

Averaging the resulting probability over the two computational inputs gives

$$\bar{L}^{(1)} = \frac{1}{2}\sum_{j=0,1} |A_{2j}^{(1)}|^2 = \frac{\lambda^2}{8}\sum_{j=0,1} |\ell_j(\nu)|^2, \tag{7}$$

where the leakage moments are

$$\ell_j(\nu) \equiv \int_0^1 \mathrm{d}s\; u^*(s)\, e^{\mathrm{i}\nu s}\, \langle 1|\tilde{U}_q(s)j\rangle. \tag{8}$$

Proposition 1 (Nominal first-order leakage closure)**.** The first-order leakage amplitude vanishes for every computational input if and only if

$$\ell_0(\nu) = 0, \qquad \ell_1(\nu) = 0. \tag{9}$$

These are two complex equations, hence four independent real constraints.

The cheaper proxy $\int_0^1 u(s)e^{\mathrm{i}\nu s}\,\mathrm{d}s = 0$ forces a notch in the bare envelope spectrum. It is not equivalent to (**??**), because the integrands contain the evolving factor $\langle 1|\tilde{U}_q(s)j\rangle$. That factor is available from the ideal qubit propagator already computed by SCQC, so the stronger condition does not require a qutrit simulation.

If a physical parameter q is uncertain, first-order insensitivity of the leakage cancellation adds

$$\partial_q \ell_0(\nu) = 0, \qquad \partial_q \ell_1(\nu) = 0, \tag{10}$$

another four real constraints for each such parameter. The important choices are anharmonicity, a common amplitude scale, and a static detuning.

3 DRAG as a post-processor

3.1 Fixed-axis transformation

Suppose first that the intended envelope $E_x(t)$ is real and a second quadrature $E_y(t)$ is available for correction. Following Motzoi et al. [**?**], remove the direct leakage coupling by the near-identity transformation

$$V(t) = \exp\left\{-\mathrm{i}\frac{E_x(t)}{2\alpha}(Y_{01} + \lambda Y_{12})\right\}. \tag{11}$$

The adiabaticity parameter is E_x/α. To first order in that parameter, the transformed Hamiltonian contains the inertial term

$$\frac{1}{2}\left(E_y + \frac{\dot{E}_x}{\alpha}\right)(Y_{01} + \lambda Y_{12}) \tag{12}$$

and an ac-Stark shift of $|1\rangle$ proportional to

$$\delta - \frac{(\lambda^2 - 4)E_x^2}{4\alpha}. \tag{13}$$

Thus the first-order DRAG controls are

$$E_y(t) = -\frac{\dot{E}_x(t)}{\alpha}, \qquad \delta(t) = \frac{(\lambda^2 - 4)E_x(t)^2}{4\alpha}. \tag{14}$$

For a transmon, $\lambda^2 = 2$, so the detuning simplifies to $\delta \approx -E_x^2/(2\alpha)$. Higher-order terms in Ref. [**?**] correct the in-phase envelope, the derivative quadrature, and the detuning at orders E_x^3/α^2, $E_x^2\dot{E}_x/\alpha^3$, and E_x^4/α^3. They are useful once η is no longer small, but they are not required for the first-order theory.

The frame transformation has no net effect on the gate precisely when

$$V(0) = V(T) = I. \tag{15}$$

For (**??**), this is the endpoint condition $E_x(0) = E_x(T) = 0$.

3.2 Complex envelope

A general SCQC pulse already uses both quadratures. The covariant extension of (**??**) is

$$\varepsilon_D(t) = \varepsilon(t) - \mathrm{i}\frac{\beta}{\alpha}\dot{\varepsilon}(t), \tag{16}$$

with a dimensionless calibration parameter β equal to one in the analytic first-order theory. In normalized variables,

$$u_D(s) = u(s) - \mathrm{i}\frac{\beta}{\nu}u'(s). \tag{17}$$

Equivalently, in Cartesian components,

$$\omega_{x,D} = \omega_x + \frac{\beta}{\nu}\omega_y', \qquad \omega_{y,D} = \omega_y - \frac{\beta}{\nu}\omega_x'. \tag{18}$$

If $u = \omega e^{\mathrm{i}\phi}$, the same correction is

$$u_D(s) = e^{\mathrm{i}\phi(s)}\left[\omega(s) + \frac{\beta}{\nu}\omega(s)\phi'(s) - \mathrm{i}\frac{\beta}{\nu}\omega'(s)\right]. \tag{19}$$

The term proportional to $\omega\phi'$ is a radial correction generated by phase modulation. It is absent from textbook fixed-axis DRAG and should not be dropped for an SCQC pulse. Derivatives should be taken on the Cartesian quadratures: near a signed-envelope zero, ω and ϕ separately can have spurious singularities even though ω_x and ω_y are smooth.

Proposition 2 (Bare spectral notch)**.** Suppose $\varepsilon(0) = \varepsilon(T) = 0$. Then

$$\int_0^T \left(\varepsilon(t) - \mathrm{i}\frac{\dot{\varepsilon}(t)}{\alpha}\right) e^{\mathrm{i}\alpha t}\,\mathrm{d}t = 0. \tag{20}$$

Proof. Integrate the derivative term by parts. The boundary term is $(\varepsilon(T)e^{\mathrm{i}\alpha T} - \varepsilon(0))/\alpha = 0$, while $-\mathrm{i}\alpha \int \varepsilon e^{\mathrm{i}\alpha t}\,\mathrm{d}t$ cancels the remaining integral of $\varepsilon e^{\mathrm{i}\alpha t}$. □

Proposition **??** does not imply Proposition **??**. The latter weights $\varepsilon^*(t)$ by $\langle 1|U_q(t)j\rangle$, and DRAG also changes U_q at order η. Equation (**??**) should therefore be read as the appropriate post-processing ansatz, followed by a small calibration of β, an overall amplitude, and a constant detuning.

3.3 Boundary conditions and calibration

The following conditions are listed in the order they become necessary.

1. $u(0) = u(1) = 0$ makes the first-order DRAG frame an identity at both endpoints. BARQ already encourages this through its vanishing-envelope construction, but a loaded or rescaled pulse should be checked.
2. $u'(0) = u'(1) = 0$ makes the added derivative quadrature start and end continuously.
3. If corrections through the mth derivative are used, the derivatives of orders $0, \ldots, m$ should vanish at both endpoints. Otherwise integration by parts leaves boundary kicks in the leakage amplitude.
4. After constructing u_D, at least the global amplitude and a constant phase or detuning offset should be recalibrated. The derivative term does not cancel every Stark and higher-order coherent error.

A practical post-processing sequence is:

1. Design $u(s)$ and $d(s)$ with ordinary BARQ and record the original closure and tantrix-area diagnostics.
2. Apply (**??**) and the Stark correction from (**??**), transformed into normalized units.
3. On a qutrit model, tune β, the amplitude scale, and a constant detuning to minimize $1-F_{\mathrm{avg}}$ and the leakage L defined in Section **??**.
4. Recompute noise generators with the corrected propagator, rather than reusing the original curve tests.
5. Repeat under variation of α, gain, timing, sampling, and the control-line transfer function.

4 Robustness after DRAG

4.1 Generator conditions

The original geometric tests are statements about a qubit propagator and a qubit noise operator. Both change once the pulse is corrected and embedded in a qutrit. Let $U_D(t)$ be the ideal qutrit evolution generated by the corrected controls. For a parameter q_a with Hamiltonian derivative $V_a(t) = \partial H/\partial q_a$, the first Magnus generator is

$$G_a = \int_0^T \mathrm{d}t\; U_D^\dagger(t)\, V_a(t)\, U_D(t). \tag{21}$$

Its computational and leakage blocks impose different requirements.

Definition 1 (First-order qutrit robustness)**.** The corrected gate is first-order robust against q_a when

$$PG_aP - \frac{1}{2}\,\mathrm{Tr}(PG_aP)\,P = 0, \qquad QG_aP = 0. \tag{22}$$

The first equation removes the traceless logical rotation. The second prevents the noise from creating leakage to first order.

For a genuine qubit model one has $P = I$, and the first equation reduces to the usual closure condition. Applying the old closure or tantrix test to the uncorrected curve is not sufficient, because neither U_D nor V_a is the operator assumed by that test.

4.2 Perturbation of dephasing closure

Let U_0 be the computational evolution before the DRAG correction and write

$$U_D = U_0 \exp\!\big[-\mathrm{i}\eta K_1 + \mathcal{O}(\eta^2)\big]. \tag{23}$$

For quasistatic detuning noise, take the interaction-picture operator $n_0(t) = U_0^\dagger(t)\sigma_z U_0(t)$ and the associated error curve $r_0(T) = \int_0^T n_0(t)\,\mathrm{d}t$. Expanding (**??**) gives

$$r_D(T) = r_0(T) + \mathrm{i}\eta \int_0^T [K_1(t), n_0(t)]\,\mathrm{d}t + \mathcal{O}(\eta^2). \tag{24}$$

Ordinary SCQC can enforce $r_0(T) = 0$. It does not enforce the commutator integral. Consequently, a pulse that is first-order dephasing-robust before post-processing generically acquires a residual generator of order η.

For a qubit, a vanishing first-order generator changes the leading coherent infidelity from quadratic to quartic in the noise strength. After unconstrained post-processing the scaling is instead

$$\begin{aligned} \text{original closure:} \quad & 1 - F = \mathcal{O}\big((\delta_z T)^4\big), \\ \text{post-processed DRAG:} \quad & 1 - F = \mathcal{O}\big((\delta_z T)^2\eta^2\big) + \mathcal{O}\big((\delta_z T)^4\big). \end{aligned} \tag{25}$$

The mixed term dominates the original quartic protection whenever $|\delta_z T| \lesssim \eta$. This is a scaling statement, not a claim that the numerical coefficient is large. Restoring the quartic regime requires the additional vector condition

$$\int_0^T [K_1(t), n_0(t)]\,\mathrm{d}t = 0, \tag{26}$$

or, more directly, the block conditions (**??**) evaluated on the final pulse.

4.3 Amplitude, DRAG gain, and anharmonicity

If one calibration error a multiplies both applied quadratures, its derivative is built from the corrected envelope,

$$V_a(t) = \frac{1}{2}\big(\Omega_{x,D}(t)X + \Omega_{y,D}(t)Y\big), \tag{27}$$

and robustness is $G_a \propto I$ on the computational block. The original tantrix-area condition is the special case of this statement before DRAG and before the qutrit embedding. If the derivative channel is calibrated independently, β must be treated as its own parameter and G_β must vanish separately. Anharmonicity uncertainty is qualitatively different: it moves the leakage phase $e^{\mathrm{i}\nu s}$ and the coefficient $1/\nu$ in (**??**) simultaneously. A first-order robust design therefore penalizes $\partial_\alpha M$, where $M = PU_DP$ is the final computational block.

Second-order cancellation follows from the next Magnus term. For two error channels,

$$G_{ab}^{(2)} = -\frac{\mathrm{i}}{2}\int_0^T \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \; [V_a^I(t_1), V_b^I(t_2)], \tag{28}$$

where the superscript I denotes the interaction picture generated by U_D. The computational block of $G_{ab}^{(2)}$, with its trace removed, is the appropriate second-order condition. In the original qubit dephasing model this is the geometric zero-area constraint.

Table 1: Noise channels that should be recomputed after DRAG.

Uncertainty	Generator	New risk
Static detuning	$\partial H/\partial\delta$	Closure shifted at $\mathcal{O}(\eta)$
Common I/Q gain	$\partial H/\partial a$	Derivative terms scale with the gain
DRAG coefficient	$\partial H/\partial\beta$	Separate derivative-channel error
Anharmonicity	$\partial H/\partial\alpha$	Notch frequency and $1/\alpha$ both move
Transfer function	filtered functional derivative	Differentiation amplifies distortion

5 Leakage-aware BARQ

Post-processing takes a curve designed for a qubit and changes its pulse. Leakage-aware BARQ instead varies the free Bézier points, and optionally a few gate-fixing parameters, under a qutrit objective. Three nested models are useful.

5.1 Spectral and perturbative objectives

The least expensive differentiable objective uses Proposition **??**:

$$J_{\mathrm{spec}} = \sum_{j=0,1} |\ell_j(\nu)|^2. \tag{29}$$

The bare-envelope proxy $J_{\mathrm{bare}} = |\int_0^1 u(s)e^{\mathrm{i}\nu s}\,\mathrm{d}s|^2$ is cheaper but weaker. The corresponding leakage probability from (**??**) is

$$J_{\mathrm{leak},1} = \frac{\lambda^2}{8}(|\ell_0|^2 + |\ell_1|^2). \tag{30}$$

Both can be evaluated from the normalized envelope and the ideal qubit propagator. Their gradients exist once the cumulative phase and arclength are computed by a differentiable quadrature rather than by a black-box integrator.

5.2 Exact qutrit objective

Let U be the full qutrit propagator and $M = PUP$ its 2×2 computational block. For a pure input drawn uniformly from the qubit Bloch sphere, the average probability of leaving that subspace is

$$L = 1 - \frac{1}{2}\,\mathrm{Tr}(M^\dagger M). \tag{31}$$

If U_g is the target qubit unitary, the Haar-averaged state fidelity is

$$F_{\mathrm{avg}} = \frac{\mathrm{Tr}(M^\dagger M) + |\mathrm{Tr}(U_g^\dagger M)|^2}{6}. \tag{32}$$

Equation (**??**) follows from

$$\int |\langle\psi|A|\psi\rangle|^2 \, \mathrm{d}\psi = \frac{|\mathrm{Tr}\, A|^2 + \mathrm{Tr}(A^\dagger A)}{d(d+1)} \tag{33}$$

with $d = 2$ and $A = U_g^\dagger M$. The leakage term (**??**) is already present in F_{avg}, but retaining it separately allows leakage to be weighted more heavily than a coherent logical rotation. Both quantities remain meaningful when M is not unitary.

5.3 What remains analytic

BARQ's point-gate-fixing construction fixes the ideal qubit operation and the endpoint contribution to dephasing closure before numerical optimization. On a qutrit, Stark shifts and virtual transitions modify M, so those constraints no longer guarantee the physical target exactly. The useful hybrid is:

- retain point-gate fixing, so the zeroth-order qubit gate and curve closure remain exact;
- constrain enough endpoint control points that u and u' vanish at $s = 0, 1$;
- optimize interior points against F_{avg}, L, and the corrected generators (**??**);
- release a global amplitude and a constant detuning during the final stage, since qutrit gate fixing is no longer purely analytic;
- test four- and five-level truncations for short gates, where two-photon and higher-state transitions are outside (**??**).

A concrete staged objective is

$$\begin{aligned} J = {} & w_g(1 - F_{\mathrm{avg}}) + w_L L + w_z \|PG_zP - \tfrac{1}{2}\,\mathrm{Tr}(PG_zP)P\|_F^2 + w_{zL}\|QG_zP\|_F^2 \\ & + w_a \|PG_aP - \tfrac{1}{2}\,\mathrm{Tr}(PG_aP)P\|_F^2 + w_{aL}\|QG_aP\|_F^2 + w_\alpha \|\partial_\alpha M\|_F^2 \\ & + w_b \int_0^1 |u'(s)|^2 \, \mathrm{d}s + w_{bb} \int_0^1 |u''(s)|^2 \, \mathrm{d}s + w_{\max} \max_s |u(s)| \\ & + w_{\mathrm{end}} \sum_{k=0,1} \sum_{s_\star \in \{0,1\}} |u^{(k)}(s_\star)|^2. \end{aligned} \tag{34}$$

The weights should be introduced in stages: target and original robustness, then nominal leakage, then parameter uncertainty, and finally bandwidth and amplitude. Turning every term on simultaneously makes the trade-off difficult to diagnose.

6 Further extensions

6.1 Choosing the gate duration

For a fixed normalized BARQ waveform, the first-order leakage depends on duration only through $\nu = \alpha T$. A coherent-error search can therefore minimize $L_1(\nu) = J_{\mathrm{leak},1}$ over T, subject to the hardware bound on peak amplitude. This is stronger than placing a zero of the bare spectrum at α, because ℓ_j contains $\tilde{U}_q$. Decoherence selects a finite optimum of a model such as

$$E(T) \approx L_{\mathrm{leak}}(\alpha T) + \frac{c_1 T}{T_1} + \frac{c_\phi T}{T_\phi} + E_{\mathrm{control}}(T). \tag{35}$$

The coefficients c_1 and c_ϕ depend on how the target gate and noise channel average relaxation and pure dephasing; they should come from the open-system simulation rather than from a universal rule.

6.2 Endpoint smoothness

Let $I(\nu) = \int_0^1 g(s)e^{\mathrm{i}\nu s}\,\mathrm{d}s$. If g and its first $m-1$ derivatives vanish at both endpoints, repeated integration by parts yields

$$I(\nu) = \left(-\frac{1}{\mathrm{i}\nu}\right)^m \int_0^1 g^{(m)}(s)e^{\mathrm{i}\nu s}\,\mathrm{d}s, \tag{36}$$

and therefore $|I| = \mathcal{O}(|\nu|^{-m})$. For leakage, $g(s) = u^*(s)\langle 1|\tilde{U}_q(s)j\rangle$. Making only u smooth does not automatically make every derivative of this product vanish, but it improves the decay. The quadratic penalties on u' and u'' in (**??**) suppress bandwidth even when an endpoint derivative cannot be constrained exactly.

6.3 Higher-derivative and multi-transition DRAG

A linear differential filter can place spectral zeros at several unwanted frequencies Δ_k:

$$u_{\mathrm{corr}}(t) = \prod_k \left(1 - \frac{\mathrm{i}}{\Delta_k}\frac{\mathrm{d}}{\mathrm{d}t}\right) u_0(t). \tag{37}$$

Under the Fourier convention $\int u(t)e^{\mathrm{i}\omega t}\,\mathrm{d}t$, each factor contributes $1 - \omega/\Delta_k$. Expanding (**??**) produces the higher-derivative DRAG family. The same construction suppresses several internal or spectator transitions, provided the corresponding endpoint derivatives vanish and the coefficients are recalibrated on the full propagator. At very short duration, direct and multiphoton transitions beyond $|2\rangle$ can dominate the residual error, so the filter frequencies must include those transitions or the simulation must retain more levels.

6.4 Fourier notches and Wah-Wah modulation

The derivative expansion is not the only way to impose the linear constraints $\ell_j(\nu_k) = 0$. A finite spectral ansatz $u(s) = \sum_n c_n f_n(s)$ converts each notch into a linear condition on the coefficients c_n. Those conditions can be combined with the BARQ gate area, endpoint, and robustness constraints. Wah-Wah control is the crowded-spectrum version of this idea: sideband modulation places a notch near a neighbour's leakage transition, while a DRAG quadrature suppresses the addressed qubit's own $1 \leftrightarrow 2$ transition [**?**, **?**]. It is relevant when a spectator transition is closer to the drive than the addressed qubit's anharmonicity.

6.5 Relation to multilevel SCQC

The generalized geometric formalism represents the interaction-picture error operator by coordinates

$$q_a(t) = \frac{1}{2}\,\mathrm{Tr}\big[\Lambda_a U^\dagger(t) Q U(t)\big], \tag{38}$$

where $\{\Lambda_a\}$ is a generator basis normalized by $\mathrm{Tr}(\Lambda_a\Lambda_b) = 2\delta_{ab}$ [**?**, **?**]. Leading-order cancellation means that every coordinate curve closes, $\int_0^T q_a(t)\,\mathrm{d}t = 0$. A generic qutrit error gives an eight-dimensional curve. Generalized curvatures can be connected recursively to the controls, but a transmon does not supply enough independent controls to realize an arbitrary curve: two quadratures and a detuning act with linked $0 \leftrightarrow 1$ and $1 \leftrightarrow 2$ matrix elements.

The perturbative intermediate theory avoids that inverse problem. Leakage lives in the four-real-dimensional space $Q\mathcal{H}P \cong \mathbb{C}^2$, and the two components of its integrated coordinate are exactly (ℓ_0, ℓ_1). Closure of this leakage curve is Proposition **??**. Thus leakage-aware BARQ adds a second, four-dimensional closure condition without abandoning the existing SU(2) curve used to design the computational gate.

7 Expected improvement and required evidence

For a fixed rotation angle, the drive amplitude scales as $1/T$. The dimensionless expansion parameter is therefore $\eta \sim 1/(|\alpha|T)$, up to a pulse-shape coefficient. A generic off-resonant leakage amplitude begins at $\mathcal{O}(\eta)$, so its probability begins at $\mathcal{O}(\eta^2)$. First-order DRAG is constructed to cancel that amplitude. The residual coherent leakage is then commonly $\mathcal{O}(\eta^2)$ in amplitude and $\mathcal{O}(\eta^4)$ in probability, before decoherence and model error are added. Table **??** gives these powers with unit prefactors. It is a regime guide, not a predicted infidelity.

Table 2: Unit-prefactor scaling for $|\alpha|/2\pi = 200\,\mathrm{MHz}$. The entries are η^2 and η^4, not calculated gate errors.

T (ns)	$\lvert\nu\rvert = \lvert\alpha\rvert T$	η	$\mathcal{O}(\eta^2)$	$\mathcal{O}(\eta^4)$
6.25	7.85	0.127	1.62×10^{-2}	2.63×10^{-4}
10	12.57	0.0796	6.33×10^{-3}	4.01×10^{-5}
20	25.13	0.0398	1.58×10^{-3}	2.51×10^{-6}
40	50.27	0.0199	3.96×10^{-4}	1.57×10^{-7}

Published results show that the attainable improvement depends strongly on duration and pulse family. Motzoi et al. obtain reductions from roughly one to several orders of magnitude in simulation, depending on the waveform [**?**]. Early transmon experiments demonstrated derivative-of-Gaussian DRAG at few-nanosecond duration [**?**]. At 4.16 ns, closed-loop optimization around DRAG reduced leakage from 0.29% to 0.044% and raised the average Clifford fidelity from 99.11% to 99.76% on that device [**?**]. Analytical FAST and higher-derivative DRAG have subsequently reached leakage below 3×10^{-5} at 6.25 ns, about a twentyfold reduction relative to a compared cosine-DRAG pulse [**?**]. None of these values can be transferred directly to a BARQ waveform: its spectrum, phase excursions, anharmonicity, and coherence environment are different. Experimental SCQC benchmarking also identifies leakage as a gate-time-dependent limitation and selects duration by balancing it against decoherence [**?**].

A numerical claim for this package needs the comparisons in Table **??**. The minimum publication-quality comparison is a qutrit simulation of the original BARQ pulse, the post-processed pulse, and a pulse reoptimized with (**??**), all evaluated at the same peak amplitude and coherence times.

Table 3: Simulation campaign needed before quoting an improvement.

Sweep	Quantities	Purpose
T and α	F_{avg}, L, logical error	Locate leakage minima and the perturbative regime
β and Stark coefficient	two-dimensional error surface	Separate leakage cancellation from phase correction
Static δ_z and gain	infidelity heat maps	Test whether robustness survives post-processing
Three to five levels	convergence of F_{avg} and L	Detect higher-level and multiphoton leakage
T_1, T_ϕ	open-system process fidelity	Choose a useful duration
Sampling and transfer function	filtered-pulse fidelity	Test sensitivity of derivative controls
α and gain mismatch	worst-case and average fidelity	Quantify calibration robustness

8 Conclusion

DRAG fits SCQC as a correction in the pulse domain. For a complex BARQ envelope the first-order post-processor is (**??**), supplemented by a quadratic Stark detuning and by the endpoint conditions $u = u' = 0$. This exactly notches the bare envelope spectrum, but it only approximately preserves the computational gate and the original closure condition. The missing robustness requirement is (**??**), evaluated with the corrected qutrit evolution; its first correction beyond ordinary curve closure is the mixed integral (**??**).

Leakage-aware BARQ imposes the two complex constraints $\ell_0(\alpha T) = \ell_1(\alpha T) = 0$ and can include their derivatives with respect to uncertain parameters. Exact qutrit propagation remains the acceptance test, with L and F_{avg} given by (**??**) and (**??**). Gate-time selection, endpoint smoothness, higher-derivative filters, and spectral notches are compatible extensions. A full eight-dimensional SU(3) construction is consistent with the general geometric formalism, whereas the practical and geometrically faithful intermediate step is closure of the perturbative leakage curve in $\mathbb{C}^2$.

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